AIE23145_LAB4

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1 Lab Assignment 4: Frame-wise Analysis of Speech Signal

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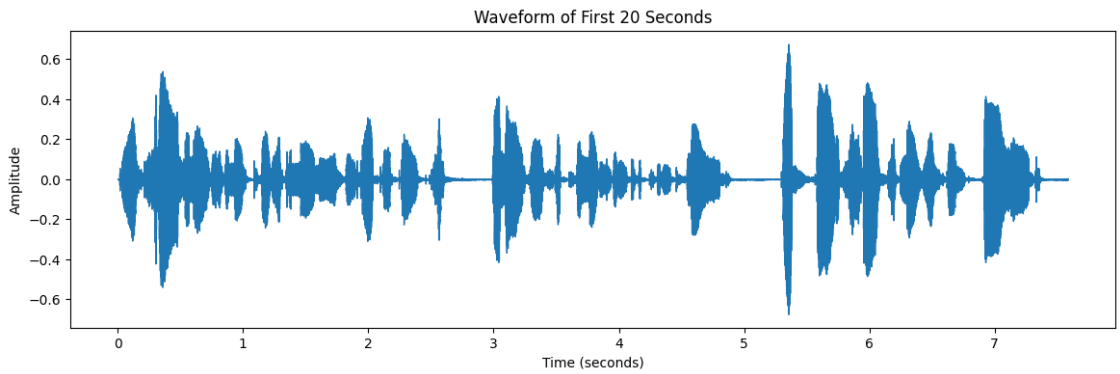
[1]:
```
import numpy as np
import matplotlib.pyplot as plt
import librosa
import librosa.display
import warnings
warnings.filterwarnings('ignore')
```

Task 1: Short-time processing of speech

[2]:
```
audio_path = "test.wav"
waveform, sr = librosa.load(audio_path, sr=16000, duration=20)

plt.figure(figsize=(14, 4))
librosa.display.waveshow(waveform, sr=sr)
plt.title("Waveform of First 20 Seconds")
plt.xlabel("Time (seconds)")
plt.ylabel("Amplitude")
plt.show()

print(f"Sampling Rate: {sr} Hz")
print(f"Duration: {len(waveform)/sr:.2f} seconds")
```




```
Sampling Rate: 16000 Hz
Duration: 7.58 seconds
```

[3]:
```
frame_length_ms = 25
frame_shift_ms = 10

frame_length = int(frame_length_ms * sr / 1000)
frame_shift = int(frame_shift_ms * sr / 1000)

num_frames = int((len(waveform) - frame_length) / frame_shift) + 1

print(f"Frame Length: {frame_length} samples")
print(f"Frame Shift: {frame_shift} samples")
print(f"Total Frames: {num_frames}")
```

```
Frame Length: 400 samples
Frame Shift: 160 samples
Total Frames: 756
```

[4]:
```
frames = np.zeros((num_frames, frame_length))

for i in range(num_frames):
    start = i * frame_shift
    end = start + frame_length
    frames[i] = waveform[start:end]

print(f"Frame Matrix Shape: {frames.shape}")
```

```
Frame Matrix Shape: (756, 400)
```

[5]:
```
def extract_features(frames, window):
    windowed = frames * window

    STE = np.sum(windowed**2, axis=1)
    STM = np.sum(np.abs(windowed), axis=1)
```

```
    ZCR = np.zeros(len(windowed))
    autocorr = []
    AMDF = []
    AMSDF = []

    for i, frame in enumerate(windowed):
        ZCR[i] = np.sum(np.abs(np.diff(np.sign(frame)))) / (2 * len(frame))

        ac = np.correlate(frame, frame, mode='full')
        ac = ac[len(ac)//2:]
        autocorr.append(ac)

        amdf_vals = []
        amsdf_vals = []
        for k in range(len(frame)):
            diff = frame[k:] - frame[:len(frame)-k]
            amdf_vals.append(np.mean(np.abs(diff)))
            amsdf_vals.append(np.mean(diff**2))

        AMDF.append(amdf_vals)
        AMSDF.append(amsdf_vals)

    return STE, STM, ZCR, np.array(autocorr), np.array(AMDF), np.array(AMSDF)

hamming_window = np.hamming(frame_length)
STE_ham, STM_ham, ZCR_ham, AC_ham, AMDF_ham, AMSDF_ham =␣
 ↪extract_features(frames, hamming_window)

print("Hamming Window Features:")
print(f"STE: {STE_ham.shape}, STM: {STM_ham.shape}, ZCR: {ZCR_ham.shape}")
print(f"Autocorr: {AC_ham.shape}, AMDF: {AMDF_ham.shape}, AMSDF: {AMSDF_ham.
 ↪shape}")
```

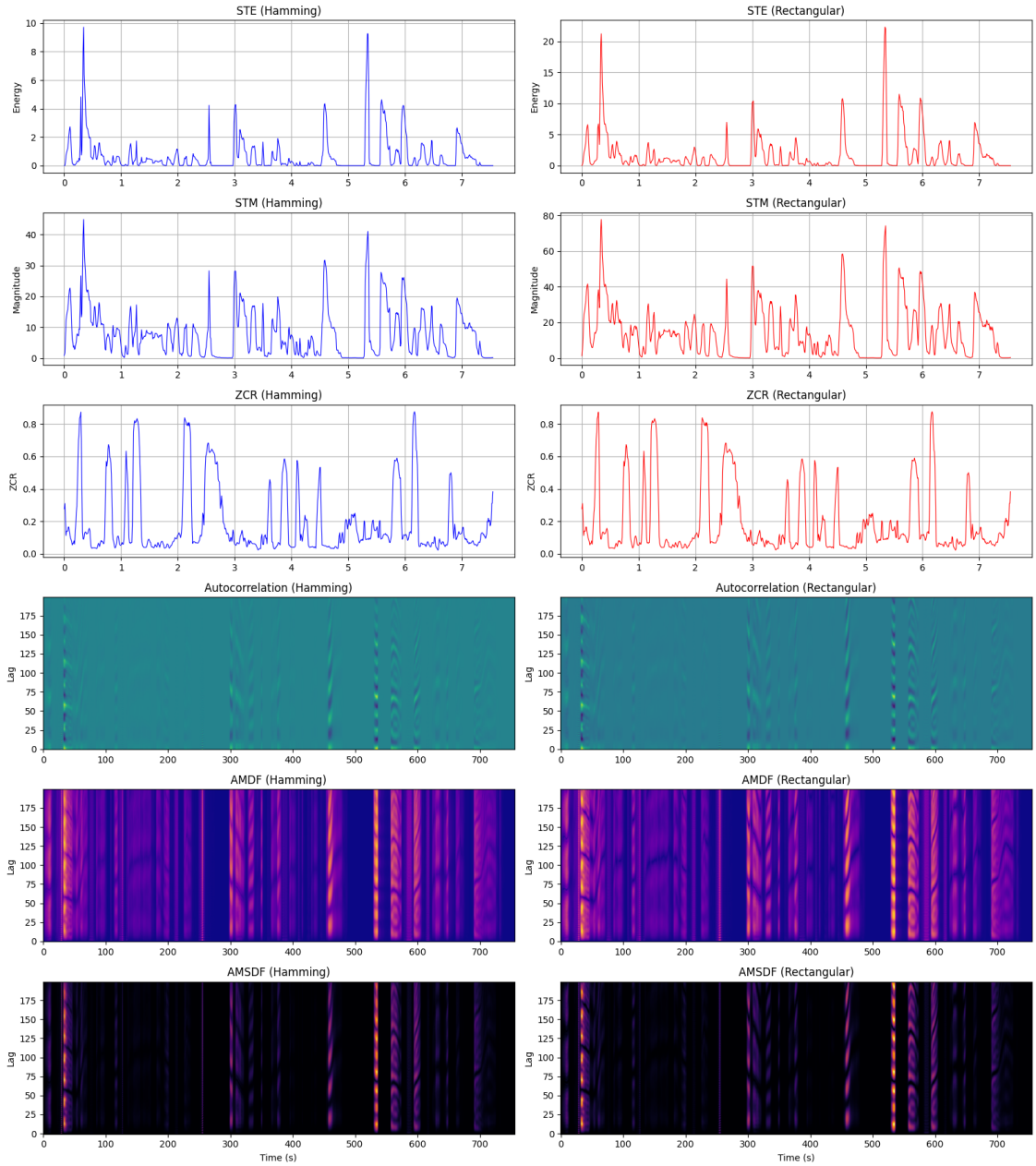
```
Hamming Window Features:
STE: (756,), STM: (756,), ZCR: (756,)
Autocorr: (756, 400), AMDF: (756, 400), AMSDF: (756, 400)
```

[6]:
```
rectangular_window = np.ones(frame_length)
STE_rect, STM_rect, ZCR_rect, AC_rect, AMDF_rect, AMSDF_rect =␣
 ↪extract_features(frames, rectangular_window)

print("Rectangular Window Features:")
print(f"STE: {STE_rect.shape}, STM: {STM_rect.shape}, ZCR: {ZCR_rect.shape}")
```

```
Rectangular Window Features:
STE: (756,), STM: (756,), ZCR: (756,)
```

[7]:
```
frame_time = np.arange(num_frames) * frame_shift / sr

fig, axes = plt.subplots(6, 2, figsize=(16, 18))

axes[0, 0].plot(frame_time, STE_ham, linewidth=0.8, color='blue')
axes[0, 0].set_title('STE (Hamming)')
axes[0, 0].set_ylabel('Energy')
axes[0, 0].grid(True)

axes[0, 1].plot(frame_time, STE_rect, linewidth=0.8, color='red')
axes[0, 1].set_title('STE (Rectangular)')
axes[0, 1].set_ylabel('Energy')
axes[0, 1].grid(True)

axes[1, 0].plot(frame_time, STM_ham, linewidth=0.8, color='blue')
axes[1, 0].set_title('STM (Hamming)')
axes[1, 0].set_ylabel('Magnitude')
axes[1, 0].grid(True)

axes[1, 1].plot(frame_time, STM_rect, linewidth=0.8, color='red')
axes[1, 1].set_title('STM (Rectangular)')
axes[1, 1].set_ylabel('Magnitude')
axes[1, 1].grid(True)

axes[2, 0].plot(frame_time, ZCR_ham, linewidth=0.8, color='blue')
axes[2, 0].set_title('ZCR (Hamming)')
axes[2, 0].set_ylabel('ZCR')
axes[2, 0].grid(True)

axes[2, 1].plot(frame_time, ZCR_rect, linewidth=0.8, color='red')
axes[2, 1].set_title('ZCR (Rectangular)')
axes[2, 1].set_ylabel('ZCR')
axes[2, 1].grid(True)

axes[3, 0].imshow(AC_ham[:, :200].T, aspect='auto', origin='lower',␣
 ↪cmap='viridis')
axes[3, 0].set_title('Autocorrelation (Hamming)')
axes[3, 0].set_ylabel('Lag')

axes[3, 1].imshow(AC_rect[:, :200].T, aspect='auto', origin='lower',␣
 ↪cmap='viridis')
axes[3, 1].set_title('Autocorrelation (Rectangular)')
axes[3, 1].set_ylabel('Lag')

axes[4, 0].imshow(AMDF_ham[:, :200].T, aspect='auto', origin='lower',␣
 ↪cmap='plasma')
axes[4, 0].set_title('AMDF (Hamming)')
```

```
axes[4, 0].set_ylabel('Lag')

axes[4, 1].imshow(AMDF_rect[:, :200].T, aspect='auto', origin='lower',␣
 ↪cmap='plasma')
axes[4, 1].set_title('AMDF (Rectangular)')
axes[4, 1].set_ylabel('Lag')

axes[5, 0].imshow(AMSDF_ham[:, :200].T, aspect='auto', origin='lower',␣
 ↪cmap='inferno')
axes[5, 0].set_title('AMSDF (Hamming)')
axes[5, 0].set_xlabel('Time (s)')
axes[5, 0].set_ylabel('Lag')

axes[5, 1].imshow(AMSDF_rect[:, :200].T, aspect='auto', origin='lower',␣
 ↪cmap='inferno')
axes[5, 1].set_title('AMSDF (Rectangular)')
axes[5, 1].set_xlabel('Time (s)')
axes[5, 1].set_ylabel('Lag')

plt.tight_layout()
plt.show()
```



Analysis:

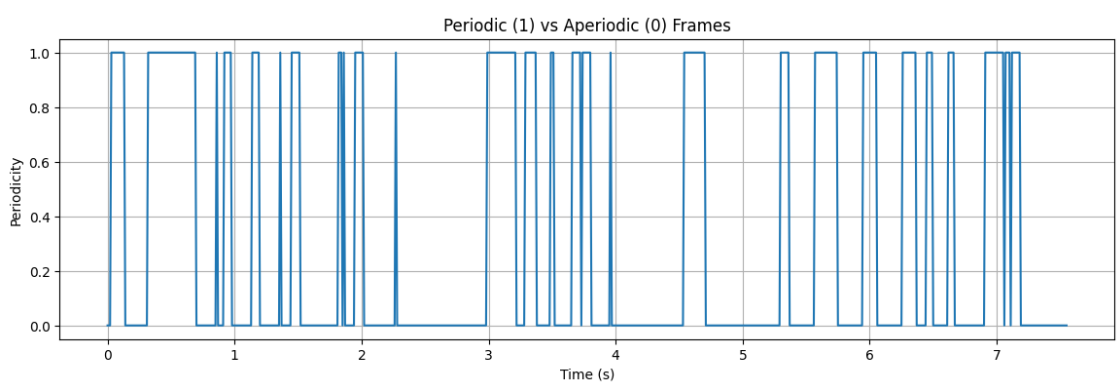
Voiced speech shows high energy, low ZCR, strong periodicity peaks. Unvoiced speech shows low energy, high ZCR, weak periodicity. Hamming window provides smoother features and better discrimination than rectangular window.

Task 2: Periodicity Analysis

[8]:
```
energy_threshold = np.percentile(STE_ham, 30)
zcr_threshold = np.percentile(ZCR_ham, 70)
```

```
periodic_frames = []

for i in range(num_frames):
    ac_frame = AC_ham[i, 20:200]
    max_peak = np.max(ac_frame)

    is_periodic = (STE_ham[i] > energy_threshold and
                   ZCR_ham[i] < zcr_threshold and
                   max_peak > 0.3)

    periodic_frames.append(1 if is_periodic else 0)

periodic_frames = np.array(periodic_frames)

print(f"Total Frames: {num_frames}")
print(f"Periodic Frames: {np.sum(periodic_frames)}")
print(f"Aperiodic Frames: {num_frames - np.sum(periodic_frames)}")

plt.figure(figsize=(14, 4))
plt.plot(frame_time, periodic_frames)
plt.title("Periodic (1) vs Aperiodic (0) Frames")
plt.xlabel("Time (s)")
plt.ylabel("Periodicity")
plt.grid(True)
plt.show()
```

```
Total Frames: 756
Periodic Frames: 232
Aperiodic Frames: 524
```



Periodicity Reflection:

Autocorrelation exhibits peaks at pitch period for voiced speech. AMDF and AMSDF show minima at pitch period. Aperiodic frames lack these patterns. These complementary measures reliably

distinguish voiced from unvoiced segments.

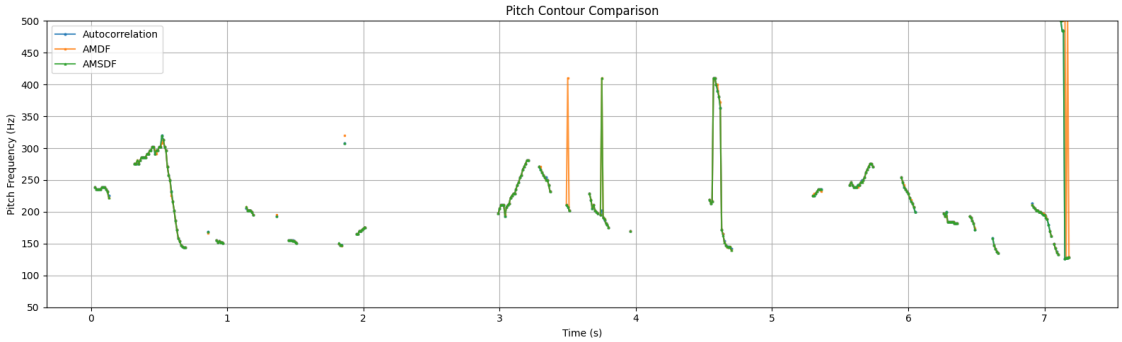
Task 3: Pitch Estimation

[9]:
```
def estimate_pitch(autocorr, amdf, amsdf, sr, min_lag=20, max_lag=200):
    ac_region = autocorr[min_lag:max_lag]
    peak_lag = np.argmax(ac_region) + min_lag
    pitch_ac = sr / peak_lag if np.max(ac_region) > 0.3 else 0

    amdf_region = amdf[min_lag:max_lag]
    min_lag_amdf = np.argmin(amdf_region) + min_lag
    pitch_amdf = sr / min_lag_amdf if np.min(amdf_region) < 0.7 * np.
 ↪mean(amdf_region) else 0

    amsdf_region = amsdf[min_lag:max_lag]
    min_lag_amsdf = np.argmin(amsdf_region) + min_lag
    pitch_amsdf = sr / min_lag_amsdf if np.min(amsdf_region) < 0.7 * np.
 ↪mean(amsdf_region) else 0

    return pitch_ac, pitch_amdf, pitch_amsdf

pitch_ac = []
pitch_amdf = []
pitch_amsdf = []

for i in range(num_frames):
    if periodic_frames[i]:
        p_ac, p_amdf, p_amsdf = estimate_pitch(AC_ham[i], AMDF_ham[i],␣
 ↪AMSDF_ham[i], sr)
        pitch_ac.append(p_ac)
        pitch_amdf.append(p_amdf)
        pitch_amsdf.append(p_amsdf)
    else:
        pitch_ac.append(0)
        pitch_amdf.append(0)
        pitch_amsdf.append(0)

pitch_ac = np.array(pitch_ac)
pitch_amdf = np.array(pitch_amdf)
pitch_amsdf = np.array(pitch_amsdf)

valid_ac = pitch_ac[pitch_ac > 0]
valid_amdf = pitch_amdf[pitch_amdf > 0]
valid_amsdf = pitch_amsdf[pitch_amsdf > 0]

print("Pitch Statistics:")
print(f"Autocorrelation - Mean: {np.mean(valid_ac):.2f} Hz, Range: {np.
 ↪min(valid_ac):.2f}-{np.max(valid_ac):.2f} Hz")
```

```
print(f"AMDF - Mean: {np.mean(valid_amdf):.2f} Hz, Range: {np.min(valid_amdf):.
↪2f}-{np.max(valid_amdf):.2f} Hz")
print(f"AMSDF - Mean: {np.mean(valid_amsdf):.2f} Hz, Range: {np.
↪min(valid_amsdf):.2f}-{np.max(valid_amsdf):.2f} Hz")
```

```
Pitch Statistics:
Autocorrelation - Mean: 222.40 Hz, Range: 125.98-516.13 Hz
AMDF - Mean: 228.54 Hz, Range: 128.00-533.33 Hz
AMSDF - Mean: 223.24 Hz, Range: 125.98-516.13 Hz
```

[10]:
```
pitch_ac_plot = np.copy(pitch_ac)
pitch_amdf_plot = np.copy(pitch_amdf)
pitch_amsdf_plot = np.copy(pitch_amsdf)

pitch_ac_plot[pitch_ac_plot == 0] = np.nan
pitch_amdf_plot[pitch_amdf_plot == 0] = np.nan
pitch_amsdf_plot[pitch_amsdf_plot == 0] = np.nan

fig, ax = plt.subplots(1, 1, figsize=(16, 5))

ax.plot(frame_time, pitch_ac_plot, linewidth=1.5, marker='o', markersize=2,␣
↪label='Autocorrelation', alpha=0.8)
ax.plot(frame_time, pitch_amdf_plot, linewidth=1.5, marker='s', markersize=2,␣
↪label='AMDF', alpha=0.8)
ax.plot(frame_time, pitch_amsdf_plot, linewidth=1.5, marker='^', markersize=2,␣
↪label='AMSDF', alpha=0.8)

ax.set_title('Pitch Contour Comparison')
ax.set_xlabel('Time (s)')
ax.set_ylabel('Pitch Frequency (Hz)')
ax.set_ylim([50, 500])
ax.legend()
ax.grid(True)

plt.tight_layout()
plt.show()
```



3.1 Task 4: Discussion

3.1.1 a) Compare voiced and unvoiced speech in terms of energy, ZCR, and periodicity

Voiced speech has high energy, low ZCR, and strong periodicity. Unvoiced speech has low energy, high ZCR, and no periodicity.

3.1.2 b) Comment on the pitch estimation results

All three methods produce similar pitch contours in voiced regions. Autocorrelation is most robust, while AMDF and AMSDF are more efficient with comparable accuracy.

3.1.3 c) Comment on the reliability of each time-domain feature for pitch detection

Energy and ZCR are reliable for voiced/unvoiced classification. Autocorrelation is highly reliable for pitch detection. AMDF and AMSDF offer good accuracy with better efficiency.